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Module 1: Password Security and Traffic Visibility

Course: Introduction to Cybersecurity

Date: March 16, 2026

Environment: Local extension + Wireshark + Microsoft Edge browser

PRACTICAL WORK REPORT

1. Introduction

This report presents the practical work of Module 1, which focuses on two fundamentals: password quality and visibility of network traffic. The goal is to link user behavior to measurable risks, and then identify effective protections (strong passwords and TLS encryption).

2. Work Environment

Component: Host Station

Detail: Local Computer (Physical PC)

Component: Network Analysis Tool

Detail: Wireshark

Component: Browser

Detail: Microsoft Edge

Component: Connection

Detail: Internet access to generate realistic traffic

3. Part 1: Password Security

Objective:

Evaluate the difference in risk between a weak and a strong password.

Method used:

Conceptual analysis of entropy and probable attacks (dictionary, brute force, reuse).

Results Achieved:

Observation: a weak password can be guessed quickly and exposes several accounts if reused.

Examples of strong passwords offered

- S3cur3! Netw0rk2026
- Cyb3r\$3cur1tyLab
- P@ck3tAn@lyz3r2026

Observation:

Strong passwords combine length, complexity, and uniqueness. It is the first line of defence against credential theft.

4. Part 2: Network Traffic Capture (Wireshark)

Tool used:

Wireshark to observe DNS, HTTP, and TLS traffic.

Approach:

1. Launch Wireshark and select the active interface (Wi-Fi).
2. Start capturing and navigate to public sites.
3. Visit google.com, wikipedia.org and bit.bf.
4. Let the capture run for about 2 minutes, then stop it.

Results Achieved:

Total number of packets observed: 68,511 (capture summary).

Main protocols: DNS, HTTP and TLS.

Observation:

Network traffic is easily observed on a local computer. Only encryption protects the content of communications.

5. Packet Analysis

DNS traffic

DNS queries show the domains you are looking for (e.g., google.com). This reveals the browsing intentions.

Screenshot:

No.	Time	Source	Destination	Protocol	Length	Info
4594	37.168868700	10.36.199.234	10.36.199.163	DNS	73	Standard query 0x454b A ncc.avast.com
4595	37.586268900	10.36.199.163	10.36.199.234	DNS	502	Standard query response 0x454b A ncc.avast.com CNAME ncc.avast.com.edgesuite.net CNAME a1488.dscd.akamai.net A 2.19.54.81 A 2.19.54.81
4605	38.182649300	10.36.199.234	10.36.199.163	DNS	74	Standard query 0x72c3 A www.google.com
4606	38.183241400	10.36.199.234	10.36.199.163	DNS	74	Standard query 0xc939 A www.google.com
4610	38.190674400	10.36.199.163	10.36.199.234	DNS	202	Standard query response 0x72c3 A www.google.com A 142.251.150.119 A 142.251.151.119 A 142.251.157.119 A 142.251.156.119 A 142.251.155.119
4611	38.190674400	10.36.199.163	10.36.199.234	DNS	202	Standard query response 0xc939 A www.google.com A 142.251.155.119 A 142.251.150.119 A 142.251.151.119 A 142.251.157.119 A 142.251.156.119
4616	38.385928300	10.36.199.234	10.36.199.163	DNS	70	Standard query 0x1b8a A c.pki.goog
4625	38.777197700	10.36.199.163	10.36.199.234	DNS	369	Standard query response 0x1b8a A c.pki.goog CNAME pki-goog.l.google.com A 142.251.142.131 NS ns2.google.com NS ns3.google.com NS ns1.google.com
4747	39.2637748500	10.36.199.234	10.36.199.163	DNS	74	Standard query 0x0b31 A x1.c.lencr.org
4753	39.279373500	10.36.199.163	10.36.199.234	DNS	503	Standard query response 0x0b31 A x1.c.lencr.org CNAME crl.root-x1.letsencrypt.org.edgekey.net CNAME e8652.dscx.akamaiedge.net A 2.19.54.81
4754	39.281422400	10.36.199.234	10.36.199.163	DNS	84	Standard query 0xe8bd HTTPS www.googleadservices.com
4755	39.284114300	10.36.199.234	10.36.199.163	DNS	84	Standard query 0x1855 A www.googleadservices.com
4757	39.293503500	10.36.199.163	10.36.199.234	DNS	336	Standard query response 0xe8bd HTTPS www.googleadservices.com HTTPS www.googleadservices.com A 142.250.185.2 NS ns4.google.com NS ns3.google.com NS ns1.google.com NS ns2.google.com
4758	39.298887900	10.36.199.163	10.36.199.234	DNS	355	Standard query response 0x1855 A www.googleadservices.com A 142.250.185.2 NS ns4.google.com NS ns3.google.com NS ns1.google.com NS ns2.google.com
4771	39.478626600	10.36.199.234	10.36.199.163	DNS	80	Standard query 0x2544 A tunnel.googlezip.net
4774	39.479962900	10.36.199.163	10.36.199.234	DNS	96	Standard query response 0x2544 A tunnel.googlezip.net A 216.239.34.157
4795	39.669800400	10.36.199.234	10.36.199.163	DNS	84	Standard query 0xbf41 HTTPS www.googletagmanager.com
4797	39.671969000	10.36.199.234	10.36.199.163	DNS	84	Standard query 0xfc6c A www.googletagmanager.com
4805	39.688848700	10.36.199.163	10.36.199.234	DNS	141	Standard query response 0xbf41 HTTPS www.googletagmanager.com SOA ns1.google.com
4807	39.689190000	10.36.199.163	10.36.199.234	DNS	355	Standard query response 0xfc6c A www.googletagmanager.com A 172.217.20.232 NS ns4.google.com NS ns1.google.com NS ns2.google.com NS ns3.google.com
4813	39.736068000	10.36.199.234	10.36.199.163	DNS	77	Standard query 0x16f6 A crl3.digicert.com
4818	39.752024600	10.36.199.163	10.36.199.234	DNS	182	Standard query response 0x16f6 A crl3.digicert.com CNAME crl.edge.digicert.com CNAME ocsd.digicert.com.cdn.cloudflare.net A 172.217.20.232
4832	39.856482200	10.36.199.234	10.36.199.163	DNS	80	Standard query 0x2b6d A tunnel.googlezip.net
4833	39.861043200	10.36.199.163	10.36.199.234	DNS	96	Standard query response 0x2b6d A tunnel.googlezip.net A 216.239.34.157
4886	40.259725000	10.36.199.234	10.36.199.163	DNS	80	Standard query 0x5f9d A tunnel.googlezip.net
4887	40.264239700	10.36.199.163	10.36.199.234	DNS	96	Standard query response 0x5f9d A tunnel.googlezip.net A 216.239.34.157
4941	40.669518500	10.36.199.234	10.36.199.163	DNS	80	Standard query 0xc949 A tunnel.googlezip.net
4942	40.673806900	10.36.199.163	10.36.199.234	DNS	96	Standard query response 0xc949 A tunnel.googlezip.net A 216.239.34.157
5164	41.625737400	10.36.199.234	10.36.199.163	DNS	72	Standard query 0x8851 A crl.pki.goog
5167	41.640233100	10.36.199.163	10.36.199.234	DNS	371	Standard query response 0x8851 A crl.pki.goog CNAME pki-goog.l.google.com A 142.251.142.131 NS ns3.google.com NS ns1.google.com NS ns2.google.com
5397	43.488495400	10.36.199.234	10.36.199.163	DNS	86	Standard query 0x3762 HTTPS encrypted-tbn3.gstatic.com
5399	43.491641700	10.36.199.234	10.36.199.163	DNS	86	Standard query 0xc584 A encrypted-tbn3.gstatic.com
5403	43.500881800	10.36.199.163	10.36.199.234	DNS	143	Standard query response 0x3762 HTTPS encrypted-tbn3.gstatic.com SOA ns1.google.com
5406	43.506036400	10.36.199.163	10.36.199.234	DNS	357	Standard query response 0xc584 A encrypted-tbn3.gstatic.com A 142.251.142.142 NS ns4.google.com NS ns3.google.com NS ns1.google.com NS ns2.google.com
5447	43.673034100	10.36.199.234	10.36.199.163	DNS	80	Standard query 0x79d5 A tunnel.googlezip.net
5451	43.678535900	10.36.199.163	10.36.199.234	DNS	96	Standard query response 0x79d5 A tunnel.googlezip.net A 216.239.34.157
5556	44.050985700	10.36.199.234	10.36.199.163	DNS	80	Standard query 0x4b8b A tunnel.googlezip.net
5557	44.056203200	10.36.199.163	10.36.199.234	DNS	96	Standard query response 0x4b8b A tunnel.googlezip.net A 216.239.34.157

HTTP traffic

HTTP is unencrypted. Paths, headers, and data can be intercepted over an untrusted network.

Screenshot:

*Wi-Fi

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htp

No.	Time	Source	Destination	Protocol	Length	Info
4599	39.888567500	10.36.199.234	2.19.54.81	HTTP	136	GET /ncc.txt HTTP/1.1
4603	38.091777600	10.36.199.234	2.19.54.81	HTTP	205	HTTP/1.1 200 OK (text/html)
4689	38.971059700	10.36.199.234	142.251.142.131	HTTP	254	GET /r/r1.cr1 HTTP/1.1
4737	39.147323200	142.251.142.131	10.36.199.234	HTTP	277	HTTP/1.1 304 Not Modified
4781	39.495352900	10.36.199.234	184.28.84.204	HTTP	281	GET / HTTP/1.1
4811	39.697826600	184.28.84.204	10.36.199.234	HTTP	317	HTTP/1.1 304 Not Modified
4968	40.917259300	10.36.199.234	172.66.2.5	HTTP	337	GET /DigicertTrustedG4CodeSigningRSA4096SHA3842021CA1.cr1 HTTP/1.1
5040	41.070936100	172.66.2.5	10.36.199.234	HTTP	389	HTTP/1.1 304 Not Modified
5053	41.116007500	10.36.199.234	172.66.2.5	HTTP	316	GET /DigicertHighAssuranceEVRootCA.cr1 HTTP/1.1
5106	41.369349700	172.66.2.5	10.36.199.234	HTTP	387	HTTP/1.1 304 Not Modified
5108	41.405838300	10.36.199.234	172.66.2.5	HTTP	310	GET /DigicertAssuredIDRootCA.cr1 HTTP/1.1
5155	41.577891400	172.66.2.5	10.36.199.234	HTTP	387	HTTP/1.1 304 Not Modified
5197	41.820365600	10.36.199.234	142.251.142.131	HTTP	261	GET /gsr1/gsr1.cr1 HTTP/1.1
5219	41.989735300	142.251.142.131	10.36.199.234	HTTP	276	HTTP/1.1 304 Not Modified
5224	42.016530300	10.36.199.234	172.66.2.5	HTTP	307	GET /DigicertGlobalRootCA.cr1 HTTP/1.1
5245	42.168982000	172.66.2.5	10.36.199.234	HTTP	387	HTTP/1.1 304 Not Modified
5247	42.198575100	10.36.199.234	172.66.2.5	HTTP	308	GET /DigicertGlobalRootG2.cr1 HTTP/1.1
5250	42.351404400	172.66.2.5	10.36.199.234	HTTP	387	HTTP/1.1 304 Not Modified
5252	42.395529800	10.36.199.234	142.251.142.131	HTTP	256	GET /r/gsr1.cr1 HTTP/1.1
5264	42.561876200	142.251.142.131	10.36.199.234	HTTP	277	HTTP/1.1 304 Not Modified
5265	42.601541400	10.36.199.234	172.66.2.5	HTTP	307	GET /DigicertGlobalRootG3.cr1 HTTP/1.1
5389	43.312026100	10.36.199.234	142.251.142.131	HTTP	254	GET /r/r4.cr1 HTTP/1.1
5396	43.481391900	142.251.142.131	10.36.199.234	HTTP	277	HTTP/1.1 304 Not Modified
5420	43.521713800	10.36.199.234	172.66.2.5	HTTP	308	GET /DigicertTrustedRootG4.cr1 HTTP/1.1
5448	43.673425100	172.66.2.5	10.36.199.234	HTTP	387	HTTP/1.1 304 Not Modified
15332	90.027476900	10.36.199.234	184.28.84.204	HTTP	281	GET / HTTP/1.1
15346	90.194150500	184.28.84.204	10.36.199.234	HTTP	317	HTTP/1.1 304 Not Modified
15387	90.392077300	10.36.199.234	172.66.2.5	HTTP	337	GET /DigicertTrustedG4CodeSigningRSA4096SHA3842021CA1.cr1 HTTP/1.1
15402	90.502680100	172.66.2.5	10.36.199.234	HTTP	389	HTTP/1.1 304 Not Modified
15406	90.540787700	10.36.199.234	172.66.2.5	HTTP	317	GET /DigicertHighAssuranceEVRootCA.cr1 HTTP/1.1
15481	90.677165500	172.66.2.5	10.36.199.234	HTTP	387	HTTP/1.1 304 Not Modified
15491	90.703246900	10.36.199.234	172.66.2.5	HTTP	311	GET /DigicertAssuredIDRootCA.cr1 HTTP/1.1
15535	90.809385000	172.66.2.5	10.36.199.234	HTTP	387	HTTP/1.1 304 Not Modified
15603	91.065062100	10.36.199.234	77.246.57.97	HTTP	336	GET /msdownload/update/v3/static/trusted/en/authorstl.cab?c31e5d09dc3fcacc HTTP/1.1
15766	91.803145000	77.246.57.97	10.36.199.234	HTTP	400	HTTP/1.1 304 Not Modified
15772	91.844713000	10.36.199.234	172.66.2.5	HTTP	307	GET /DigicertGlobalRootG2.cr1 HTTP/1.1
15799	91.955771200	172.66.2.5	10.36.199.234	HTTP	387	HTTP/1.1 304 Not Modified
15831	91.986934200	10.36.199.234	172.66.2.5	HTTP	309	GET /DigicertTrustedRootG4.cr1 HTTP/1.1

0000 96 b2 8a 29 8d 62 9e 95 d0 dd e4 6c 08 00 45 28 --- b...-1-E(

Paquets : 68511 - Affichés : 39 (0.1%) - Perdus : 0 (0.0%)

Profil : Default

TLS traffic

TLS encrypts application content. The metadata (IP, handshake) remains visible, but the data is protected.

Screenshot:

*Wi-Fi

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tls

No.	Time	Source	Destination	Protocol	Length	Info
4748	39.265081500	142.251.156.119	10.36.199.234	TLSv1.3	1466	[TCP Out-Of-Order], Application Data
4749	39.268306700	10.36.199.234	142.251.156.119	TLSv1.3	93	Application Data
4752	39.278232000	10.36.199.234	142.251.156.119	TLSv1.3	472	Application Data
4762	39.317918300	142.251.142.142	10.36.199.234	TLSv1.3	93	Application Data
4763	39.322087200	142.251.142.142	10.36.199.234	TLSv1.3	1466	[TCP Previous segment not captured], Application Data
4765	39.367026100	142.251.142.142	10.36.199.234	TLSv1.3	150	Application Data, Application Data
4766	39.367026100	142.251.142.142	10.36.199.234	TLSv1.3	93	Application Data
4776	39.485081300	142.251.156.119	10.36.199.234	TLSv1.3	85	[TCP Out-Of-Order], Application Data
4779	39.488948000	10.36.199.234	142.250.185.2	TLSv1.3	872	Client Hello (SNI=www.googleadservices.com)
4783	39.533387800	142.251.142.142	10.36.199.234	TLSv1.3	1110	[TCP Out-Of-Order], Application Data
4785	39.545732000	10.36.199.234	142.251.142.142	TLSv1.3	93	Application Data
4786	39.546622300	10.36.199.234	142.251.142.142	TLSv1.3	89	Application Data
4787	39.558826400	10.36.199.234	192.178.25.97	TLSv1.3	1112	Application Data, Application Data
4788	39.650257000	142.251.156.119	10.36.199.234	TLSv1.3	795	Application Data
4789	39.650458800	142.251.156.119	10.36.199.234	TLSv1.3	85	Application Data
4799	39.674147000	142.250.185.2	10.36.199.234	TLSv1.3	1466	Server Hello, Change Cipher Spec
4800	39.677209600	142.250.185.2	10.36.199.234	TLSv1.3	115	Application Data
4802	39.679371300	10.36.199.234	142.250.185.2	TLSv1.3	128	Change Cipher Spec, Application Data
4803	39.682712500	10.36.199.234	142.250.185.2	TLSv1.3	146	Application Data
4804	39.682855700	10.36.199.234	142.250.185.2	TLSv1.3	817	Application Data
4812	39.731087900	192.178.25.97	10.36.199.234	TLSv1.3	93	Application Data
4816	39.744102200	10.36.199.234	142.251.142.74	TLSv1.3	406	Application Data, Application Data, Application Data
4820	39.768475200	140.82.114.26	10.36.199.234	TLSv1.2	83	Application Data
4821	39.769843800	10.36.199.234	140.82.114.26	TLSv1.2	83	Ignored Unknown Record[Illegal Segments]
4823	39.809764200	208.103.161.2	10.36.199.234	TLSv1.2	108	Application Data
4824	39.811248300	10.36.199.234	208.103.161.2	TLSv1.2	112	Application Data
4827	39.844156100	142.250.185.2	10.36.199.234	TLSv1.3	1032	Application Data, Application Data
4828	39.846410000	10.36.199.234	142.250.185.2	TLSv1.3	85	Application Data
4830	39.847759100	142.250.185.2	10.36.199.234	TLSv1.3	85	Application Data
4837	39.873043900	142.250.185.2	10.36.199.234	TLSv1.3	433	Application Data
4839	39.875081000	142.250.185.2	10.36.199.234	TLSv1.3	93	[TCP Previous segment not captured], Application Data
4846	39.926765100	142.251.142.74	10.36.199.234	TLSv1.3	93	Application Data
4847	39.961373100	142.251.142.74	10.36.199.234	TLSv1.3	612	Application Data
4848	39.961373100	142.251.142.74	10.36.199.234	TLSv1.3	268	Application Data
4850	39.962646600	142.251.142.74	10.36.199.234	TLSv1.3	85	Application Data
4851	39.962646600	142.251.142.74	10.36.199.234	TLSv1.3	93	Application Data
4853	39.974789300	10.36.199.234	142.251.142.74	TLSv1.3	89	Application Data
4854	39.975961900	10.36.199.234	142.251.142.74	TLSv1.3	93	Application Data

0000 9e 95 d0 dd e4 6c 96 b2 8a 29 8d 62 08 00 45 00 ----1...-E-

Paquets : 68511 - Affichés : 33413 (48.8%) - Perdus : 0 (0.0%)

Profil : Default

6. Answers to the Analysis Questions

Question 1: How many packages were captured?

68,511 packages were captured during the session.

Question 2: Name three protocols observed.

DNS, HTTP, and TLS.

Question 3: Why is encryption important?

It prevents an attacker in a listening position from reading sensitive data (identifiers, messages, personal information).

Question 4: What information can an attacker obtain in the clear?

Identifiers, sites visited, forms entered and any information transmitted without encryption.

7. Safety Observations

Exposure of data in plain text:

HTTP traffic and some DNS requests can leak information about usage and content.

Importance of TLS

TLS greatly reduces the interception surface by encrypting application flows.

Password hygiene

Strong, unique passwords limit reuse and automated attacks.

Recommendations

- Use a password manager to generate unique secrets.

- Favor HTTPS everywhere and avoid unreliable networks without a VPN.
- Monitor connections and update applications.

8. Conclusion

This lab highlights two major risks: weak passwords and exposure of unencrypted traffic. Simple measures (strong passwords, HTTPS/TLS) significantly reduce the likelihood of attack and improve security posture.